A Real-Time Research Project Report

on

SMART WASTE MANAGEMENT GARBAGE MONITORING USING IOT

Submitted in partial fulfillment of the requirements for the award of the degree

of

Bachelor of Technology

in

CSE [Artificial Intelligence & Machine Learning]

by

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CERTIFICATE

This is to certify that the Real-Time Research Project Report entitled "**Smart Waste Management Garbage Monitoring Using IOT"** being submitted by **K Tarun [247R1A6687] N Shruthi [247R1A6698] Shyam Sunder [247R1A66B1] T Dakshinya [247R1A66B7]** in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in CSE [Artificial Intelligence & Machine Learning] to the Jawaharlal Nehru Technological University, Hyderabad is a record of Bonafide work carried out by them under my guidance and supervision during the Academic Year 2025 – 26.

The results embodied in this report have not been submitted to any other University or Institute for the award of any other degree

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ABSTRACT

Waste management has become a major challenge in modern cities due to increasing population and urbanization. Traditional garbage collection systems often lead to overflowing bins, unpleasant odours, and unhygienic surroundings. To address this issue, a **Smart Waste Management System using the Internet of Things** is proposed to monitor garbage levels in waste bins in real time.

The system uses sensors installed inside garbage bins to measure the level of waste. When the garbage reaches a certain threshold level, the sensor sends data to a microcontroller which then transmits the information to a central monitoring system through an internet connection. This allows municipal authorities to monitor the status of bins remotely and schedule garbage collection efficiently.

The proposed system helps reduce overflow of waste bins, improves cleanliness in public areas, and optimizes waste collection routes. It also reduces unnecessary fuel consumption and operational costs by ensuring that garbage is collected only when required. Overall, this system contributes to a cleaner and smarter environment and supports the development of smart cities.

LIST OF FIGURES

S. No	Figure Name	Page No
Figure 3.1.1	USB Cable	6
Figure 3.2.1	ESP – 32 Microcontroller	9
Figure 3.2.2	Ultrasonic Sensor	13
Figure 4.2	Proposed Methodology	20
Figure 4.3.1	Class Diagram for Smart Waste Management	21
Figure 4.3.2	Use Case Diagram for Smart Waste Management	22
Figure 4.3.3	Component Diagram for Smart Waste Management	23
Figure 4.3.4	Sequence Diagram for Smart Waste Management	23
Figure 4.3.5	Activity Diagram for Smart Waste Management	24

TABLE OF CONTENTS

Chapter Title	Page No
Abstract	i
List of Figures	ii
1. Introduction	
1.1 Motivation	1
1.2 Problem Statement	1
1.3 Project Objective	1
2. Literature Survey	
2.1 Traditional Waste Management System	2
2.2 Smart Waste Management Concept	2
2.3 Role of Internet Of Things	3
2.4 Sensor-Bases Garbage Level Detection	3
2.5 Wireless Communication Technologies	3
2.6 Real-Time Monitoring and Data Management	3
2.7 Advantages of IoT Based Waste Monitoring Systems	4
2.8 Future Scope	4
3. Requirement Specification	
3.1 Software Requirements	5
3.2 Hardware Requirements	9
4. System Design	
4.1 System Design	17
4.2 Proposed Methodology	20
4.3 UML Design	21
4.3.1 Class Diagram	21
4.3.2 Use Case Diagram	22

4.3.3 Component Diagram 23

4.3.4 Sequence Diagram 23

4.3.5 Activity Diagram 24

4.4 Technology Description 24

5. Implementation

5.1 Implementation 28

6. Testing

6.1 Testing Definition 31

6.2 Test Cases 31

7. Results 33

8. Conclusion 34

9. Future Scope 35

10. References 36

1.INTRODUCTION

1.1 MOTIVATION

Improper waste management is a serious problem in many cities and towns. Overflowing garbage bins, irregular waste collection, and lack of proper monitoring lead to unhygienic surroundings, bad odors, and the spread of diseases. Traditional waste collection systems depend on fixed schedules rather than the actual fill level of bins, which often results in either overflowing garbage or unnecessary collection trips.

1.2 PROBLEM STATEMENT

Improper waste disposal and inefficient garbage collection are major challenges in many urban and rural areas. Traditional waste management systems rely on fixed schedules for garbage collection without considering the actual fill level of the bins. As a result, garbage bins often overflow before the collection time, leading to foul odors, environmental pollution, and the spread of diseases.

Such a system can detect the garbage level in bins, send alerts when the bins are full, and help authorities schedule waste collection effectively, thereby improving hygiene and environmental sustainability.

1.3 PROJECT OBJECTIVE

The main objectives of the Smart Waste Management – Garbage Monitoring System using Internet of Things are:

1. To monitor garbage levels in waste bins using sensors and detect when the bin is full.
2. To send real-time information about the garbage level to a central monitoring system through the internet.
3. To prevent overflowing of garbage bins by providing timely alerts to waste collection authorities.
4. To improve efficiency in waste collection by collecting garbage only when the bins are full

2. LITERATURE SURVEY

Several researchers have studied smart waste management systems using the Internet of Things to improve garbage collection and monitoring in urban areas.

Many studies highlight that traditional waste collection systems depend on fixed schedules and manual inspection, which often leads to overflowing bins, environmental pollution, and inefficient use of resources. Researchers proposed smart garbage bins equipped with sensors that can detect the fill level of waste and send alerts to authorities when the bin is nearly full. This helps ensure timely waste collection and improves urban cleanliness.

A survey conducted by Sayali S. Chalke et al. (2018) analyzed various IoT-based garbage monitoring techniques. The study explained that increasing population and improper waste disposal create serious environmental problems. The authors suggested using sensors, wireless communication, and dashboards to monitor garbage containers and notify garbage collection trucks when bins reach a threshold level.

2.1 Traditional Waste Management Systems

Traditional waste management systems mainly depend on manual labor and fixed schedules for garbage collection. In most cities and towns, municipal workers collect waste from garbage bins at predetermined times without checking whether the bins are full or empty. This approach often results in inefficient waste management because some bins overflow before the scheduled collection time, while others may be collected even when they are not full. Overflowing garbage leads to unpleasant odors, environmental pollution, and the spread of diseases. These problems highlight the need for a smarter and more efficient waste monitoring system.

2.2 Smart Waste Management Concept

Smart waste management is an advanced approach that uses modern technologies to improve the efficiency of waste collection and disposal. Instead of relying only on manual observation, this system uses sensors and communication technologies to monitor the level of waste in garbage bins. The system can automatically notify waste management authorities when the bins are full. This ensures timely collection of garbage and helps maintain cleanliness in public places. Smart waste management systems are considered an important part of modern urban development and sustainable environmental practices.

2.3 Role of Internet of Things in Waste Management

The Internet of Things (IoT) plays a major role in the development of smart waste management systems. IoT refers to a network of interconnected devices that can communicate and exchange data through the internet. In a smart garbage monitoring system, sensors installed inside waste bins collect information about the garbage level and send the data to a microcontroller. The microcontroller processes the data and transmits it to a central server or monitoring system.

2.4 Sensor-Based Garbage Level Detection

Sensors are the key components used to detect the level of waste inside garbage bins. Ultrasonic sensors are commonly used because they can measure the distance between the sensor and the surface of the garbage. When the waste reaches a certain threshold level, the sensor sends a signal to the microcontroller indicating that the bin is full. Other sensors such as infrared sensors or weight sensors can also be used depending on the system design. Sensor-based monitoring ensures accurate detection of waste levels and helps automate the waste collection process.

2.5 Wireless Communication Technologies

Wireless communication technologies are used to transfer data from the smart garbage bins to the monitoring system. Technologies such as Wi-Fi, GSM, and other IoT communication modules enable the system to send alerts and status updates in real time. These communication technologies allow waste management authorities to access bin information from remote locations through mobile applications or web dashboards. By using wireless communication, the system reduces the need for manual inspection and improves the speed of response for garbage collection.

2.6 Real-Time Monitoring and Data Management

Real-time monitoring systems allow authorities to track the status of garbage bins continuously. The data collected by sensors is transmitted to a central monitoring platform where it is displayed in the form of graphs, notifications, or alerts. When a bin reaches its maximum capacity, the system automatically sends a notification to the waste collection team. This ensures that garbage is collected on time and prevents bins from overflowing. Real-time

Monitoring also helps in analyzing waste generation patterns, which can be useful for improving waste management strategies.

2.7 Advantages of IoT-Based Waste Monitoring Systems

IoT-based waste monitoring systems provide many advantages compared to traditional methods. These systems reduce manual effort by automating the monitoring process. They help prevent garbage overflow by sending timely alerts to waste collectors. The system also improves route planning for garbage collection vehicles, which reduces fuel consumption and operational costs. In addition, maintaining clean surroundings improves public health and enhances the overall quality of life in urban areas. Such systems also support environmental sustainability by promoting efficient waste management practices.

2.8 Future Scope

The proposed smart waste management system using the Internet of Things has significant potential for future development and improvement. As technology continues to evolve, the system can be enhanced with advanced features that improve efficiency, sustainability, and overall waste management practices.

In the future, the system can also support automatic waste segregation. Smart bins can be designed with sensors and image recognition technologies to identify different types of waste such as biodegradable waste, plastic, metal, and glass.

Finally, the smart waste management system can be implemented on a large scale in urban, semi-urban, and rural areas. By expanding the system across cities and towns, governments can improve sanitation, reduce environmental pollution, and create cleaner living spaces. With continuous technological advancements, smart waste management will play a key role in building sustainable and environmentally responsible communities.

3. REQUIREMENTS SPECIFICATION

This chapter gives an overview of the software and hardware components required for our project.

3.1 SOFTWARE REQUIREMENTS

Introduction to Arduino IDE

Arduino is a prototype platform (open-source) based on an easy-to-use hardware and software. It consists of a circuit board, which can be programmed (referred to as a microcontroller) and a ready-made software called Arduino IDE (Integrated Development Environment), which is used to write and upload the computer code to the physical board.

Key features of Arduino IDE

Arduino boards are able to read analog or digital input signals from different sensors and turn it into an output such as activating a motor, turning LED on/off, connect to the cloud and many other actions.

You can control your board functions by sending a set of instructions to the microcontroller on the board via Arduino IDE (referred to as uploading software).

Steps to upload the program

In this section, we will learn in easy steps, how to set up the Arduino IDE on our computer and prepare the board to receive the program via USB cable.

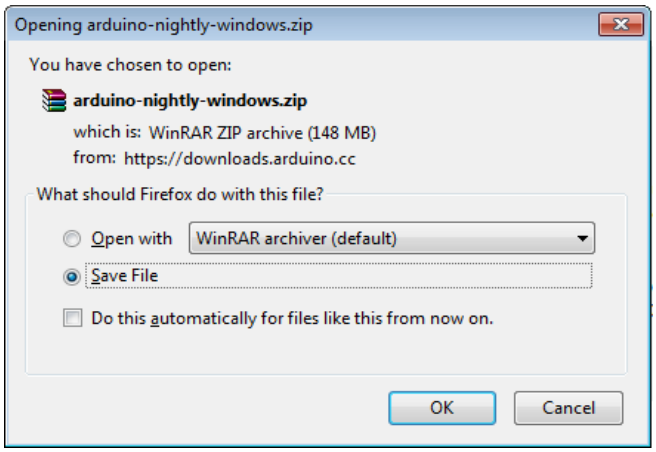
Step 1: First you must have your Arduino board (you can choose your favorite board) anda USB cable.

In case you use Arduino UNO, Arduino Duemilanove, Nano, Arduino Mega2560, or Diecimila, you will need a standard USB cable (A plug to B plug), the kind you would connect to a USB printer as shown in the following image.

Fig 3.1.1 USB Cable

Step 2: Download Arduino IDE Software.

You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.



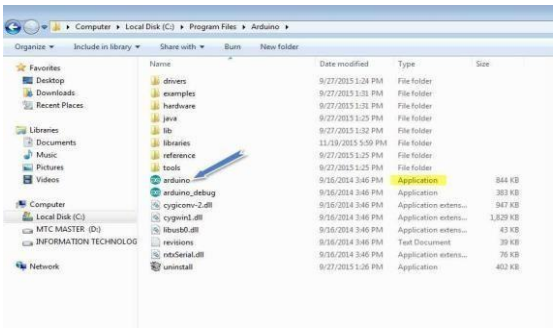
Step 3: Power up your board.

The Arduino Uno, Mega, Duemilanove and Arduino Nano automatically draw power from either, the USB connection to the computer or an external power supply. If you are using an Arduino Diecimila, you have to make sure that the board is configured to draw power from the USB connect ion.The power source is selected with a jumper,a

small piece of plastic that fits onto two of the three pins between the USB and power jacks. Check that it is on the two pins closest to the USB port. Connect the Arduino board to your computer using the USB cable. The green power LED (labeled PWR) should glow.

Step 4: Launch Arduino IDE.

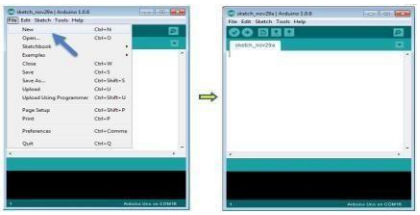
After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Doubleclick the icon to start theIDE.



Step 5: Open your first project.

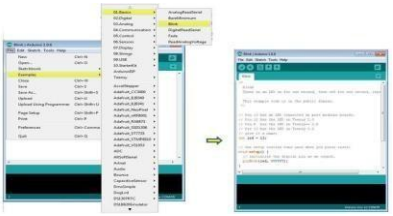
Once the software starts, you have two options: Create a new project. Open

an existing project example.



To create a new project, select File -->New.To open

To open an existing project example, select File -> Example -> Basics -> Blink.



Here, we are selecting just one of the examples with the name Blink. It turns the LED on and off with some time delay. You can select any other example from the list.

Step 6: Select your Arduino board.

To avoid any error while uploading your program to the board, you must select the correct Arduino board name, which matches with the board connected to your computer.

Go to Tools -> Board and select your board

Step 7: Select your serial port.

Select the serial device of the Arduino board. Go to Tools ->Serial Port menu. This is likely to be COM3 or higher (COM1 and COM2 are usually reserved for hardware serial ports). To find out, you can disconnect your Arduino board and re-open the menu, the entry that disappears should be of the Arduino board. Reconnect the board and select that serialport.

Step 8: Upload the program to your board.

Before explaining how we can upload our program to the board, we mustdemonstrate the function of each symbol appearing in the Arduino IDE toolbar.

3.2 HARDWARE REQUIREMENTS

ESP-32

ESP32 is the SoC (System on Chip) microcontroller which has gained massive popularity recently. Whether the popularity of ESP32 grew because of the growth of IoT or whether IoT grew because of the introduction of ESP32 is debatable.

Fig 3.2.1 ESP-32 Microcontroller

Before we delve into the actual reasons for the popularity of ESP32, let's take a look at some of its important specifications. The specs listed below belong to the ESP32 WROOM 32 variant.−

- Integrated Crystal− 40 MHz
- Module Interfaces− UART, SPI, I2C, PWM, ADC, DAC, GPIO, pulse counter, capacitive touch sensor
- Integrated SPI flash− 4 MB
- ROM− 448 KB (for booting and core functions)
- SRAM− 520 KB
- Integrated Connectivity Protocols− WiFi, Bluetooth, BLE
- On−chip sensor− Hall sensor

- Operating temperature range− −40 − 85 degrees Celsius
- Operating Voltage− 3.3V
- Operating Current− 80 mA (average)

Installation Steps

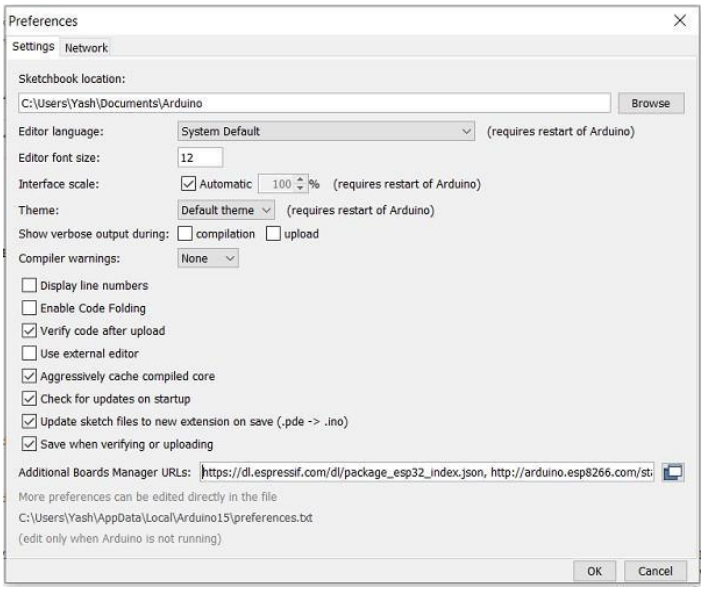
Now, to install the ESP32 board in the Arduino IDE, you need to follow the below steps −

- Make sure you have Arduino IDE (preferably the latest version) installed on your machine
- Open Arduino and go to File −> Preferences
- In the Additional Boards Manager URL, enter https://dl.espressif.com/dl/package_esp32_index.json

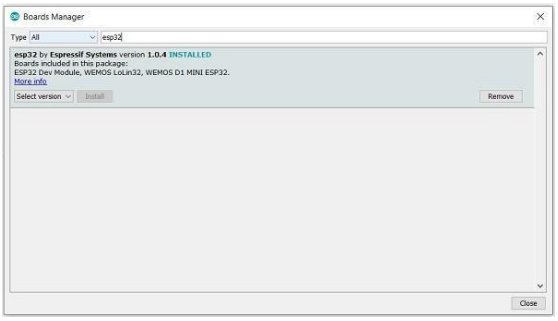
  In case you have an existing JSON file's URL in the preferences (this is likely if you've installed ESP8266, stm32duino, or any such additional board in the IDE), you can just append the above

  path to the existing path, using a comma. An example is shown below, for ESP8266 and ESP32 boards −

  http://arduino.esp8266.com/stable/package_esp8266com_index.json,
  https://dl.espressif.com/dl/package_esp32_index.json

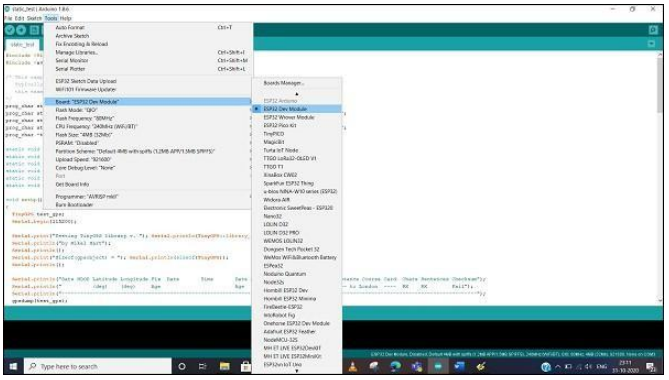


Go to Tools –> Board–> Boards Manager. A pop–up would open up. Search for ESP32 and install the **esp32** by **Espressif Systems** board. The image below shows the board already installed because I had installed the board before preparing this tutorial.

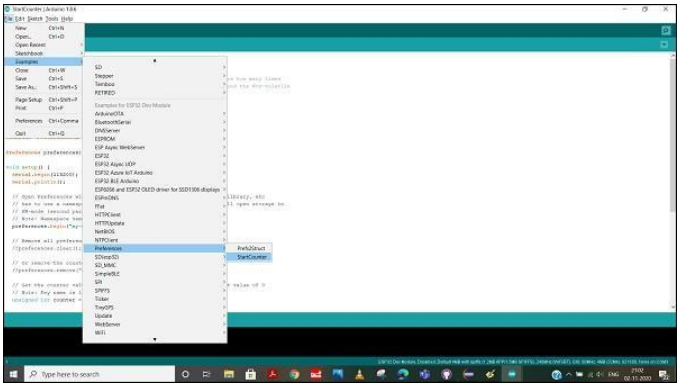


Verifying the Installation

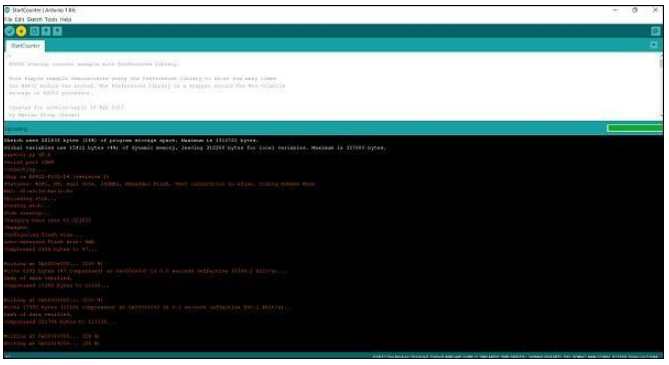
Once your ESP32 board has been installed, you can verify the installation by going to Tools Boards. You can see a whole bunch of boards under the ESP32 Arduino section. Choose the board of your choice. If you are not sure which board best represents the one you have, you can choose ESP32 Dev Module.



Next, connect your board to your machine using the USB Cable. You should see an additional COM Port under Tools−> Port. Select that additional port. In case you see multiple ports, you can disconnect the USB and see which port disappeared. That port corresponds to ESP32.



Once the port is identified, pick any one example sketch from File −> Examples. We will choose the StartCounter example from File −> Examples −> Preferences −>StartCounter. Open that sketch, compile it and flash it into the ESP32 by clicking on the Upload button (the right arrow button, besides the Compile button).



ULTRASONIC SENSOR

Then open the Serial Monitor using Tools −> Serial Monitor, or simply by pressing Ctrl + Shift + M on your keyboard. You should see the counter value getting incremented after every ESP32 restart.

Ultrasonic sensors are electronic devices that calculate the target's distance by emission of ultrasonic sound waves and convert those waves into electrical signals. The speed of emitted ultrasonic waves traveling speed is faster than the audible sound. There are mainly two essential elements which are the transmitter and receiver. Using the piezoelectric crystals, the transmitter generates sound, and from there it travels to the target and gets back to the receiver component.

To know the distance between the target and the sensor, the sensor calculates the amount of time required for sound emission to travel from transmitter to receiver. The calculation is done as follows:

D = 1/2 T * C

Where 'T' corresponds to time measured in seconds

'C' corresponds to sound speed = 343 measured in mts/sec

Ultrasonic sensor working principle is either similar to sonar or radar which evaluates the target/object attributes by understanding the received echoes from sound/radio waves correspondingly. These sensors produce high-frequency sound waves and analyse the echo which is received from the sensor. The sensors measure the time interval between transmitted and received echoes so that the distance to the target is known.

Fig 3.2.2 Ultrasonic Sensor

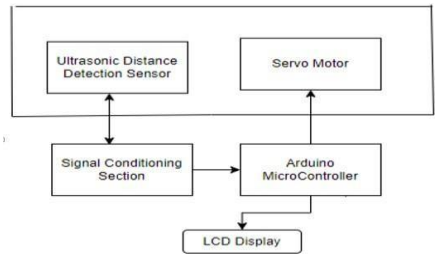
Ultrasonic Sensor Pin Diagram

VCC – This pin has to be connected to a power supply +5V.

TRIG – This pin is used to receive controlling signals from the Arduino board. This is the triggering input pin of the sensor

ECHO – This pin is used for sending signals to the Arduino board where the Arduino calculates the pulse duration to know the distance. This pin is the ECHO output of the sensor.

GND – This pin has to be connected to the ground.
The below picture shows the **ultrasonic sensor block diagram** for distance measurement.



Ultrasonic Sensor Block Diagram

LCD

LCD is a flat display technology, stands for "Liquid Crystal Display," which is generally used in computer monitors, instrument panels, cell phones, digital cameras, TVs, laptops, tablets, and calculators. It is a thin display device that offers support for large resolutions and better picture quality. The older CRT display technology has replaced by LCDs, and new display technologies like OLEDs have started to replace LCDs. An LCD display is most commonly found with Dell laptop computers and is available as an active-matrix, passive-matrix, or dual-scan display. The picture is an example of an LCD computer monitor.

The Liquid Crystal library allows you to control LCD displays that are compatible with the Hitachi HD44780 driver. There are many of them out there, and you can usually tell them by the 16-pin interface.

Output of the sketch on a 16x2 LCD

The LCDs have a parallel interface, meaning that the microcontroller has to manipulate several interface pins at once to control the display. The interface consists of the following pins:

- A **register select (RS) pin** that controls where in the LCD's memory you're writing data to. You can select either the data register, which holds what goes on the screen, or an instruction register, which is where the LCD's controller looks for instructions on what to do next.

- A **Read/Write (R/W) pin** that selects reading mode or writing mode
- An **Enable pin** that enables writing to the registers
- 8 **data pins (D0 -D7)**. The states of these pins (high or low) are the bits that you're writing to a register when you write, or the values you're reading when you read.

BUZZER

An audio signaling device like a beeper or buzzer may be electromechanical or piezoelectric or mechanical type. The main function of this is to convert the signal from audio to sound. Generally, it is powered through DC voltage and used in timers, alarm devices, printers, alarms, computers, etc. Based on the various designs, it can generate different sounds like alarm, music, bell & siren.

Fig. 3.2.3 Buzzer Pin Configuration

The pin configuration of the buzzer is shown above. It includes two pins namely positive and negative. The positive terminal of this is represented with the '+' symbol or a longer terminal. This terminal is powered through 6Volts whereas the negative terminal is represented with the '-'symbol or short terminal and it is connected to the GND terminal.

4 .SYSTEM DESIGN

4.1 System Design

The system design of the smart waste management system explains how different components work together to monitor garbage levels and send information to the monitoring system using the **Internet of Things**.

1. Overview of the System

The smart waste management system consists of the following components:

1. **Smart Garbage Bin** – Equipped with an ultrasonic sensor to detect garbage level.
2. **ESP32 Microcontroller** – Processes sensor data and communicates with the cloud server.
3. **Cloud Server / IoT Platform** – Stores and displays bin status data in real time.
4. **Monitoring Dashboard / Mobile App** – Used by authorities to check garbage levels and receive alerts.
5. **Waste Collection Team** – Receives notifications about full bins and collects garbage.

Dataflow:

Sensor → ESP32 → Cloud Server → Dashboard → Alerts → Waste Collection Team

2. Sensor Module

- **Ultrasonic Sensor (HC-SR04):**
  - Installed at the top of the garbage bin.
  - Measures the distance from the sensor to the garbage surface.
  - Converts the distance to a garbage fill level
- **Function:** Detects whether the bin is empty, partially full, or full.

3. Microcontroller Unit (ESP32)

- Reads the sensor data and processes it.
- Compares measured garbage level with predefined thresholds.
- Decides the bin status:
  - **Empty:** Garbage level below 25%
  - **Partially Full:** 25% – 85%
  - **Full:** Above 85%
- Sends the processed data to the **cloud server** via its **built-in Wi-Fi**.

4. Communication Module

- The ESP32's **Wi-Fi** module transmits data to the IoT cloud platform in real time.
- Supports multiple bins sending data simultaneously to the same cloud server.
- Optional: SMS or email notifications can also be triggered for authorities when bins are full.

5. Monitoring and Notification System

- The cloud server collects and stores data from multiple smart bins.
- Sends **alerts to the waste collection team** when a bin reaches the full threshold.
- Allows authorities to **plan optimized collection routes**, reducing fuel and time.

6. Power Supply Unit

- ESP32 and sensors are powered using a **battery or AC adapter**.
- Optional: **Solar panels** can be added to make the bins self-powered for outdoor use

7.Working Process

1. The **ultrasonic sensor** continuously measures the distance between the sensor and the garbage surface.
2. The **ESP32** reads the sensor data and calculates the fill level.
3. The ESP32 sends the data to the **cloud server**.
4. The **dashboard** updates in real time with bin status.
5. If a bin is full, the system **sends an alert** to the waste collection team.
6. The waste team collects garbage, empties the bin, and the sensor detects an empty bin, resetting the system.

4.2 PROPOSED METHODOLOGY

Fig 4.2 Proposed Methodology

4.3 UML DESIGN

Unified Modeling Language (UML) is used to design the structure and behavior of the system. The following diagrams help in visualizing the components and their interactions within the system.

4.2.1 CLASS DIAGRAM

The class diagram for the Smart Waste Management System using ESP32 consists of key classes such as WasteBin, UltrasonicSensor, ESP32Controller, IoTServer, AlertSystem, and User. The WasteBin uses an UltrasonicSensor to detect garbage level, which is processed by the ESP32Controller and sent to the IoT server. The server analyzes the data and triggers alerts through the AlertSystem when the bin is full. The User monitors the system and responds to notifications. Relationships include association, dependency, and communication between system components.

Key Points

- Class diagram shows static structure of the system
- Includes classes, attributes, and methods
- ESP32 acts as the central controller
- Ultrasonic sensor is used for level detection
- IoT cloud enables remote monitoring
- Alerts are generated when bin is full

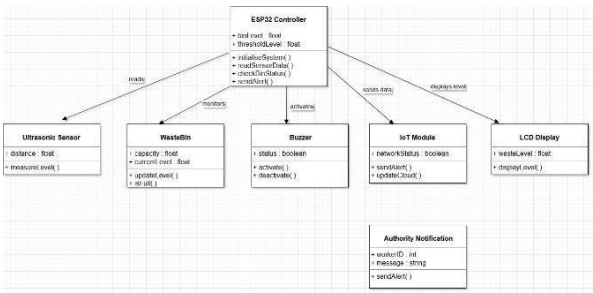


Fig 4.3.1: Class diagram

4.2.2 USE CASE DIAGRAM

The **Use Case Diagram** represents how different users interact with the smart waste management system. It identifies the **actors** involved in the system and the **functions** they can perform.

Actors

1. **Citizen/User** – People who use the smart garbage bins in public places.
2. **Municipal Authority/Admin** – The authority responsible for waste collection and monitoring.
3. **IoT System** – The hardware system consisting of ESP32, sensors, and internet connectivity.

Use Cases

- **Monitor Garbage Level:** Sensors detect the level of garbage inside the bin.
- **Send Data to Cloud:** ESP32 sends sensor data to the cloud server through WiFi.
- **View Garbage Status:** Admin can check bin status using a mobile or web application.
- **Send Notification:** The system automatically sends alerts when the bin becomes full.

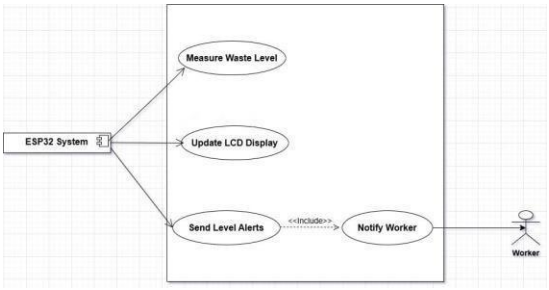


Fig 4.3.2 : Use Case Diagram

4.2.3 COMPONENT DIAGRAM

The Component Diagram represents the physical and software components and shows how these components interact with each other.

The mobile application allows the municipal authority to monitor the status of bins in real time.

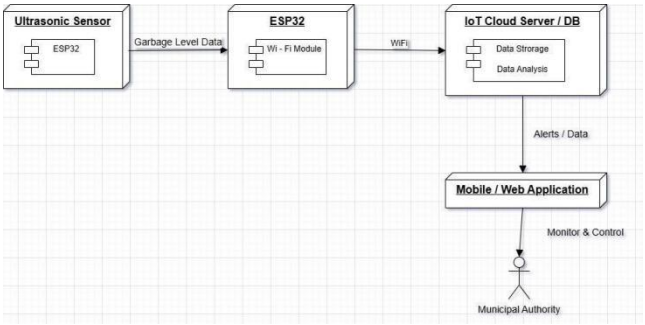


Fig 4.3.3: Component diagram

4.2.4 SEQUENCE DIAGRAM

The Sequence Diagram illustrates the step-by-step communication between system components. The working process includes

1. The ultrasonic sensor detects the garbage level in the bin.
2. The sensor sends the measured data to the ESP32 microcontroller.
3. The ESP32 processes the data and checks the garbage level and sends the data to cloud.
4. The cloud server stores and analyses the data.
5. If the garbage level crosses the threshold, the system sends a notification.
6. The municipal authority receives the alert and schedules garbage collection.

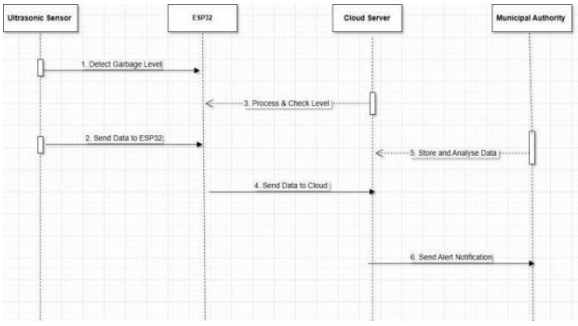


Fig 4.3.4: Sequence Diagram

4.2.5 ACTIVITY DIAGRAM

- The activity diagram shows the complete workflow of the smart waste management system.
- The process starts when the ultrasonic sensor detects the garbage level inside the bin.
- The sensor sends the measured data to the ESP32 microcontroller.
- If the bin is not full, the system continues monitoring the garbage level.
- If the bin is full, the system sends a notification to the municipal authority or waste

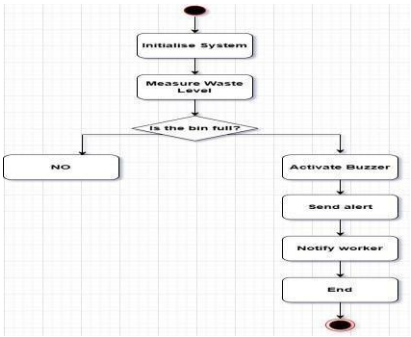


Fig 4.3.4: Activity Diagram

4.4 Technology Description

- **Dashboard/Web Application:**
  - Displays the status of all smart bins (Empty / Partial / Full).
  - Uses **color-coded indicators** for quick visual identification.
  - Allows authorities to **track bin locations**, check historical data, and plan collection routes.

- **Mobile App (Optional):**
  - Provides push notifications to waste collection teams.
  - Can include real-time GPS for bin locations and route optimization.
- **Technologies Used:** HTML, CSS, JavaScript, React.js (for web), or Flutter/React Native (for mobile apps).

2. Back-End

The back-end handles **data processing, business logic, and communication** between the ESP32 microcontroller and the front-end dashboard.

- **ESP32 Controller:**
  - Reads sensor data from the ultrasonic sensor.
  - Processes data to determine bin status (Empty / Partially Full / Full).
  - Sends data to the cloud server over Wi-Fi.
- **Server-Side Logic:**
  - Receives data from multiple ESP32 devices.
  - Stores processed data in a database.
  - Triggers alerts when bins reach the threshold.
- **Technologies Used:** Python (Flask/Django), Node.js, or PHP.

3. Database

The database stores information about each smart bin, including historical and real-time data.

- **Data Stored:**

  - Bin ID and location
  - Garbage level status
  - Timestamp of last collection
  - Alerts history
- **Database Types:**
  - **Relational Database:** MySQL, PostgreSQL for structured data.
  - **NoSQL Database:** Firebase Realtime Database or MongoDB for faster IoT data handling.

- **Purpose:** Enables analytics, historical reporting, and real-time monitoring.

4. Security

Security ensures that **data from smart bins and notifications are safe and reliable**.

- **Data Encryption:**
  - ESP32 can use HTTPS or MQTT with TLS/SSL to send data securely to the cloud.
- **Authentication:**
  - Only authorized users (municipal authorities and waste team) can access the dashboard.
- **Data Privacy:**
  - Bin locations and collection schedules are protected to prevent misuse.
- **Regular Updates:**
  - Firmware updates for ESP32 ensure the system remains secure from

vulnerabilities.

5. Notification Services

The notification service ensures timely alerts for efficient garbage collection.

- **Alert Triggers:**
    - When a bin reaches the full threshold (e.g., >75%).
- **Notification Methods:**
    - **SMS:** Using services like Twilio.
    - **Email:** Automated emails for municipal authorities.

  Push Notifications: Mobile apps alert the waste collection team in real time.
- **Purpose:**
    - Prevents overflow of garbage bins.
    - Helps authorities schedule and optimize collection routes efficiently.

5. IMPLEMENTATION

The implementation phase focuses on turning the design and methodology into a working system. The system uses ESP32 microcontroller, ultrasonic sensors, IoT cloud, and dashboard applications for real-time garbage monitoring.

5.1 Hardware Setup

1. ESP32 Microcontroller:
    - Acts as the central controller.
    - Has built-in Wi-Fi for cloud communication.
2. Ultrasonic Sensor (HC-SR04):
    - Mounted at the top of each garbage bin.
    - Measures the distance between the sensor and garbage surface to detect fill level.
3. Power Supply:
    - ESP32 and sensor powered using a battery pack or an AC adapter.
    - Optional: Solar panels can provide eco-friendly power.
4. Optional Indicators:
    - LED or buzzer installed on bins for local alerts (e.g., red LED lights up when full).

Wiring:

- Sensor pins connected to ESP32's GPIO pins.
- VCC and GND connected to the power supply.
- Ensure correct voltage levels for ESP32 (3.3V) and sensor (5V compatible).

5.2. Software Implementation

Step 1: Programming the ESP32

- Use Arduino IDE to program the ESP32.
- Code functions include:
    - Reading distance from ultrasonic sensor.
    - Calculating garbage level.
    - Comparing the level with thresholds (Empty < 25%, Partial 25–75%, Full > 75%).
    - Sending data to the cloud via Wi-Fi (MQTT or HTTP protocol).

Step 2: Cloud Setup

- Use an IoT platform (e.g., ThingSpeak, Blynk, Firebase).
- ESP32 sends garbage data to the cloud in real time.
- Cloud server stores historical data and triggers notifications when bins are full.

Step 3: Front-End Dashboard

- Web or mobile application displays:
    - Bin status with color codes (Green: Empty, Yellow: Partial, Red: Full)
    - Location of bins on a map
    - Alerts for full bins
- Dashboard can also generate reports for authorities to analyze garbage trends.

5.3. Notification Service

- The system automatically triggers notifications for the waste collection team

  - Push notifications for mobile apps
- Alerts are sent when bins cross the full threshold.
- Ensures timely garbage collection and prevents overflow.

5.4 Testing and Calibration

1. Sensor Calibration:
   - Measure empty and full bin distances and configure thresholds accordingly.
2. ESP32 Connectivity Test:
   - Ensure ESP32 successfully connects to Wi-Fi and sends data to the cloud.
3. Notification Test:
   - Simulate a full bin scenario and check if alerts reach the intended recipients.
4. End-to-End Testing:
   - Verify real-time monitoring, dashboard updates, and waste collection workflow.

5.5 Deployment.

1. Install smart bins at selected locations.
2. Mount ultrasonic sensors and connect ESP32 controllers.
3. Power the system using batteries or solar panels.
4. Connect all ESP32 devices to the IoT platform.
5. Provide dashboard and mobile app access to municipal authorities and collection teams.
6. Monitor and maintain the system periodically for optimal performance.

6. TESTING

6.1 TESTING DEFINITION:

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of tests. Each test type addresses a specific testing requirement.

6.2 TEST CASES

TC01: Sensor Distance Measurement

- **Description:** Verify ultrasonic sensor measures distances accurately.
- **Input/Scenario:** Place object at varying heights inside the bin.
- **Expected Output:** Sensor outputs correct distance values corresponding to Empty, Partial, and Full levels.

TC02: Garbage Level Calculation

- **Description:** Verify ESP32 calculates garbage fill percentage correctly.
- **Input/Scenario:** Distance data from the sensor.
- **Expected Output:** ESP32 calculates the correct % garbage fill.

TC03: Threshold Detection – Empty

- **Description:** Test bin status for empty condition.
- **Input/Scenario:** Fill level <25%.
- **Expected Output:** Bin status displayed as Empty (Green).

TC04: Threshold Detection – Partial

- **Description:** Test bin status for partially full condition.
- **Input/Scenario:** Fill level 25–75%.
- **Expected Output:** Bin status displayed as Partial (Yellow).

TC05: Threshold Detection – Full

- **Description:** Test bin status for full condition.
- **Input/Scenario:** Fill level >75%.
- **Expected Output:** Bin status displayed as Full (Red).

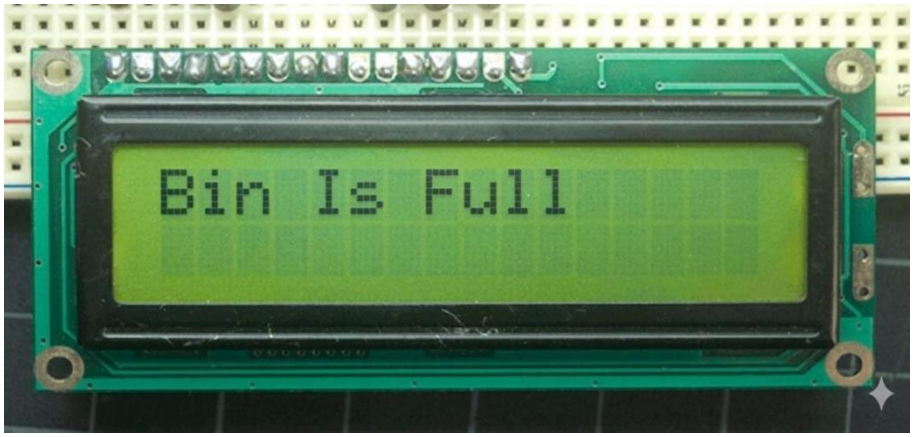
TC06: Data Transmission to Cloud

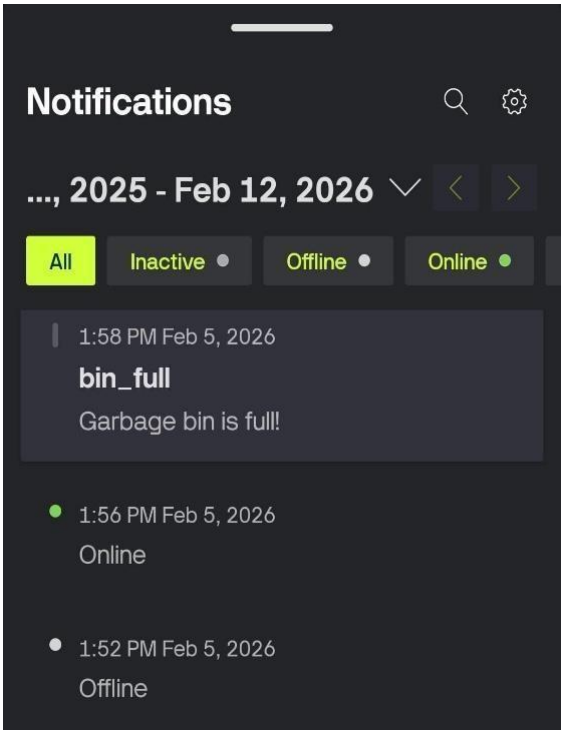
- **Description:** Verify ESP32 sends processed data to cloud server.
- **Input/Scenario:** ESP32 sends garbage level data.
- **Expected Output:** Data stored correctly in cloud server.

TC07: Dashboard Real-Time Update

- **Description:** Check if dashboard updates bin status in real-time.
- **Input/Scenario:** Cloud receives bin data.
- **Expected Output:** Dashboard updates bin status instantly.

7. RESULTS


Bin Is Full


Notifications
..., 2025 - Feb 12, 2026
All
Inactive
Offline
Online
1:58 PM Feb 5, 2026
bin_full
Garbage bin is full!
1:56 PM Feb 5, 2026
Online
1:52 PM Feb 5, 2026
Offline

8.CONCLUSION

The ESP32-based Smart Waste Management System demonstrates a modern, efficient, and sustainable approach to urban waste management by integrating IoT, sensors, cloud computing, and real-time monitoring. The project successfully achieves its core objective of providing a smart solution to monitor garbage levels in bins, ensuring that municipal authorities and waste collection teams are informed in real time, leading to optimized waste collection, reduced operational costs, and improved public hygiene.

The testing phase, including unit testing, integration testing, and acceptance testing, confirmed that the system operates seamlessly under normal and adverse conditions, such as network interruptions or power fluctuations. Sensors accurately detect garbage levels, ESP32 processes and transmits data reliably, the cloud stores and updates information in real time, and notifications are sent promptly to the concerned personnel. The dashboard provides a clear and intuitive interface, enabling authorities to plan efficient garbage collection routes, thereby minimizing overflow, odor, and unhygienic conditions.

This system has demonstrated several **key advantages**:

- **Efficiency:** Reduces unnecessary trips for garbage collection by only notifying authorities when bins are full.
- **Cost-effectiveness:** Low-cost ESP32 microcontrollers and sensors make the system affordable for large-scale deployment.
- **Scalability:** Easily accommodates multiple bins across various locations, making it suitable for urban, semi-urban, and smart city applications..

In conclusion, the ESP32-based Smart Waste Management System is not just a technological implementation but a step towards smarter, cleaner, and more sustainable cities. It proves that IoT-based smart solutions can transform traditional urban services, ensuring efficiency, reliability, and improved quality of life for citizens.

9.FUTURE SCOPE

The Smart Waste Management System using ESP32 and IoT has significant potential to evolve into a smarter, more efficient, and sustainable solution for urban waste management.

1. **AI and Analytics:** Future versions can use **machine learning** to predict garbage generation patterns and optimize collection routes, reducing fuel usage and operational costs.
2. **Automated Segregation:** Integrating sensors for **biodegradable, recyclable, and hazardous waste** can enable **automatic sorting**, improving recycling efficiency and reducing manual labor.
3. **Energy Efficiency:** Using **solar-powered bins** and low-power ESP32 modes will make the system self-sustainable and suitable for remote areas.
4. **Enhanced Notifications & Apps:** Citizen engagement through mobile apps can allow **reporting of overflowing bins**, improving participation and urban cleanliness.
5. **Smart City Integration:** The system can be connected to **municipal platforms** for centralized monitoring, better planning, and data-driven decision-making.
6. **Industrial & Hazardous Waste Management:** Expanding to factories, hospitals, and hazardous areas can improve safety and environmental compliance.
7. **Global Scalability:** With IoT protocols like **LoRaWAN** and modular design, the system can be deployed across cities worldwide, handling multiple bins efficiently.

10.REFERENCES

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